

Tariq Cisse

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ABOUT

Aspiring biostatistician broadly interested in design and analysis of clinical trials as well as longitudinal and hierarchical studies. I find myself particularly drawn to topics including targeted learning, model-robust inference and stochastic processes. Bilingual: fluent in both English and French.

EDUCATION

University of Washington

Seattle, WA

Candidate, Master of Science in Biostatistics

2025 – 2027

- Coursework in Statistical Inference, Design of Medical Studies/Clinical Trials, Longitudinal & Multilevel/Hierarchical Models, Real Analysis, Machine Learning & Statistical Computing, + more

University of Toronto

Toronto, ON

Honours Bachelor of Science in Statistics, with High Distinction

2021 – 2025

- Dean's List: 2022, 2023, 2024, 2025
- Coursework in Advanced Machine Learning and Deep Learning, Statistical Inference, Probability, Stochastic Processes, and Multivariate, Categorical, Time-Series, Regression and Survival Analysis, + more

PROFESSIONAL EXPERIENCE

Department of Biostatistics, University of Washington

Seattle, WA

Research Assistant | R

January 2026 – Present

- Under supervision of Dr. James Hughes, exploring longitudinal (GEE and mixed-effect) models and performing simulations for model-robust analysis of stepped wedge trials and estimand-based inference for treatment effect.
- Synthesizing results across 50+ studies, primarily comparing instantaneous treatment effect model with time-varying effect specifications (e.g., exposure time indicator, cubic spline) and different variance structures.
- Managing local data repository and continuously enhancing data analysis and result-processing software by leveraging **steppedwedge** R package, thus improving flexibility and reproducibility of analyses.

Sound Pharmaceuticals, Inc.

Seattle, WA

Clinical Data Coordinator (Intern) | R, Python, MATLAB, ClinCapture EDC

October 2025 – August 2026

- Working in Sound Pharmaceuticals, Inc.'s (SPI) Clinical Data Management team to support projects and studies with therapeutics for prevention and treatment of sensorineural hearing loss and tinnitus.
- Conducting statistical analyses and reports of clinical and safety endpoints as well as patient-reported outcomes from Phase I-III trials + open-label studies (completed and ongoing).
- Reviewing datasets and specifications (AE, FT, VS, etc.) with detailed reports, as well as User Acceptance Testing (UAT) of eCRFs in the EDC, in accordance with clinical protocols and CDISC standards (SDTM/ADaM).

Dalla Lana School of Public Health, University of Toronto

Toronto, ON

Research Assistant | R, Python, PowerQuery, GIS

September 2024 – April 2025

- Under supervision of Dr. Erija Ge, assisted the Geoinformatics of Spatial and Environmental Health Laboratory (Ge-iSEE) with data analysis and of population health indicators in Ontario and in the UK.
- Developed engines to increase efficiency of data cleaning, processing, reconciliation and mapping of spatial data on air pollution, vegetation health and other environmental factors at 300,000+ sites spanning 20+ years.
- Performed literature screenings and reviews for meta-analysis and systematic review, and to assess study designs and statistical modeling methods.

Financial Services Regulatory Authority of Ontario (FSRA)

Toronto, ON

Risk Analyst (Co-op) | Python, SQL Server, Power BI, PowerQuery

January – August 2023

- Raised efficiency standard in production of reports on 1,000+ Canadian employers and improved consistency of risk insights by leading data visualization processes, database management, validation and reconciliation tasks.
- Regularly provided critical insights and, performed and documented stochastic projections of fund's financial position with internal model for the production of sector reviews, risk appetite statements and inaugural report.
- Supported statistical analysis project to inform strategies with new investment manager by processing and analyzing data on various capital market variables.

RESEARCH EXPERIENCE & PROJECTS

Neo-Natal Mortality Estimation

June 2026 – Present

- Set to work with Dr. Jon Wakefield exploring methods for estimation of neo-natal mortality in the Ivory Coast with small-area estimation and spatial models in presence of sparse data.

Latent Class & Processes Estimation with Extended Mixed Models

May – June 2026

- Developed teaching materials and companion lecture recording introducing methods to classmates for estimation of latent class mixed models and latent process mixed models. Project inspired by [Proust-Lima et al., 2017](#).
- Provided simplified theoretical framework for each model and motivating examples, then outlined procedures for estimation, inference and post-fit model diagnostics.
- Demonstrated implementation of models in R using **lcmm** package and longitudinal cognitive aging dataset with supporting visualizations and interpretations.

German Breast Cancer Study Group - Survival Analysis

April 2025

- Final project for readings class in survival analysis: Investigated treatment effect of hormonal therapy on recurrence-free survival of 686 cancer patients, comparing non-parametric, semi-parametric and parametric models.
- Analyzed patients' survival data by producing Kaplan-Meier curves, Stratified Cox Proportional Hazards (PH) model and Weibull parametric model with Gamma frailty, adjusting for age, tumor size and grade, hormones, etc..
- Built and validated models using scaled Schoenfeld, Martingale and deviance residuals, log-log plots and Likelihood Ratio Tests.

Child Mind Institute: Severity Impairment Index - Ensemble Classification

March – April 2025

- Machine Learning project facilitated by the Child Mind Institute for better understanding and prediction of problematic internet use among 3,960 children and teens, and assessing of their mental health.
- Trained XGBoost, Random Forrest and Support Vector Machine (SVM) classifiers with 40 features (demographic, physical activity, mental health indicators, etc...), and using Quadratic Weighted Kappa (QWK) as metric.
- Combined all three models into a Logistic Regression meta-classifier via stacking for robustness and achieved very competitive results in comparison to top submissions in corresponding Kaggle competition (Top 8%).

Analysis of Hero Rats' Sensory Preferences - Pre-Post Bayesian Modeling

May – August 2024

- Research project supervised by Dr. So-hee Kang, modelling 32 HeroRATS' behavioral preferences for TNT and Chamomile odors based on early-age scent exposure and training for detection of landmines and TB.
- Performed Exploratory Data Analysis (EDA) including (Inter- and Intra-) Reliability analyses and PCA for operationalization of "rats' preference".
- Fit and validated Bayesian Linear Mixed Model with 500-sample Posterior Predictive Checks. The project was presented at the University of Toronto Scarborough 2024 Undergraduate Research Symposium.

TEACHING

Drop-in Tutor, University of Washington

October 2025 – June 2026

- Statistics Tutor and Study Center

PUBLICATIONS

From RA-ship at Dalla Lana:

- Kim, H., Lai, K., Cisse, T., Wei, X., & Ge, E. (2026). Association between air pollution and chronic cough: Evidence from a systematic review and meta-analysis of observational studies. *Environmental Research*, 124108. <https://doi.org/10.1016/j.envres.2026.124108>
- Grubic, N., Ge, E., Zang, Z., Lavigne, E., Campitelli, M. A., Golding, H., Cisse, T., Koh, M., Batomen, B., Liu, J., & Wei, X. (2026). Association between air temperature and COVID-19 emergency department visits in Ontario, Canada: A population-based case-crossover study. *International Journal of Hygiene and Environmental Health*, 276, 114856. <https://doi.org/10.1016/j.ijheh.2026.114856>